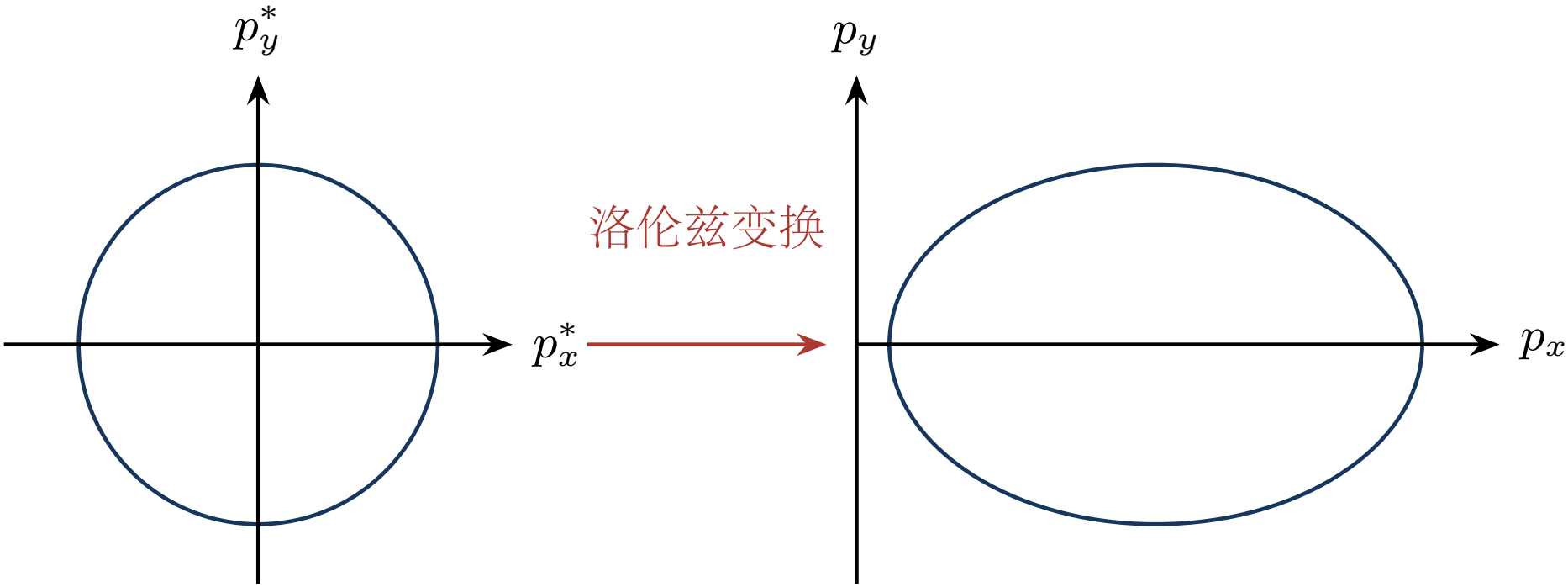

质心系：圆　　实验室系：平移、伸长

$p_x = \gamma_{\mathrm{cm}}(p^* \cos\chi + \beta_{\mathrm{cm}} E^*)$，$p_y = p^* \sin\chi$（取 $c = 1$）。